

Summary

Project Overview

This project focuses on analyzing telecom customer churn using SQL to understand why customers leave the company and which customer segments are most likely to churn. The analysis transforms raw customer data into meaningful business insights by evaluating customer demographics, subscription details, revenue, service usage, and churn behavior. The objective is to help the business identify high-risk customers and develop strategies to improve customer retention and profitability.

Customer Overview

This section provides a high-level summary of the customer base by calculating:

- Total number of customers
- Active customers
- Churned customers
- Overall churn rate

These KPIs help measure customer retention performance and provide a baseline for further analysis.

Revenue Analysis

Revenue-related metrics are analyzed to understand the financial impact of customer churn, including:

- Average monthly charges
- Total revenue generated
- Revenue lost due to churn

This analysis highlights how customer attrition directly affects business profitability and recurring revenue.

Customer Demographics Analysis

Customer churn is examined across different demographic groups such as:

- Age groups
- Gender
- Marital status

The goal is to identify customer segments that are more likely to discontinue the service, allowing the business to design targeted retention campaigns.

Service & Contract Analysis

Customer churn is compared based on service-related attributes, including:

- Internet service type

- Contract type
- Payment method

This helps determine which services or subscription plans are associated with higher churn and whether certain payment methods or contract lengths influence customer loyalty.

Geographic Analysis

Customer churn is analyzed across different cities to identify locations with the highest number of churned customers. This enables businesses to recognize regional performance differences and prioritize customer engagement in high-risk areas.

Churn Categories & Reasons

The project identifies the most common churn categories and customer-reported churn reasons. Understanding why customers leave provides valuable insights into service quality, pricing, customer support, or competitive challenges faced by the business.

Business Recommendations

Based on the analysis, the following recommendations can help reduce customer churn:

- Identify high-risk customer segments and launch targeted retention campaigns before customers leave.
- Encourage customers to switch to long-term contracts by offering discounts or loyalty benefits.
- Improve customer support for service categories with higher churn rates.
- Monitor cities with consistently high churn and investigate regional issues such as network quality or competitor presence.
- Analyze the most common churn reasons to address customer pain points through product improvements and better service quality.
- Offer personalized plans and promotional offers to customers with high monthly charges who are more likely to churn.
- Continuously monitor churn KPIs through dashboards to evaluate the effectiveness of retention strategies.

Conclusion

This project demonstrates how SQL can be used to solve real-world business problems by converting raw telecom customer data into actionable insights. The analysis supports strategic decision-making by helping businesses understand customer behavior, reduce churn, protect revenue, and improve overall customer satisfaction.